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LEONA COLLING, University of Tübingen, Tubingen, Baden-Wurttemberg, Germany

MAREIKE KHOLIN

DETMAR MEURERS, Knowledge Media Research Center, Tubingen, Germany

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A Learning Analytics Dashboard for K-12 English Teachers - Bridging the Gap Between Student Process Data and Teacher Needs

Leona Colling
leona.colling@uni-tuebingen.de
University of Tübingen
Tübingen, Germany

Mareike Kholin
kholin@die-bonn.de
German Institute for Adult Education
- Leibniz Centre for Lifelong Learning
Bonn, Germany

Detmar Meurers
d.meurers@iwm-tuebingen.de
Leibniz-Institut für Wissensmedien
(IWM)
Tübingen, Germany

ABSTRACT

Educational technologies are being used more and more in secondary school settings. This increases the amount of students' learning related data produced and stored. To keep up with this rise and to get most out of the collected data, teachers need digital tools that support and facilitate their pedagogical decision-making process. Learning analytics dashboards can be a good source to provide teachers with necessary insights into their students' learning processes. However, for such tools to be effective and actionable, they have to be aligned with teachers' needs and thus, provide and visualize data in a concise and structured way. We therefore conducted a survey study with 11 English teachers from K-12 secondary schools in Germany who evaluated the assumed usefulness of possible dashboard features. Based on these findings, we developed a teacher dashboard incorporating the most desired functionalities, such as a quickly accessible summary of strengths, weaknesses and support needs, or an overview of current misconceptions and competencies alongside additional metrics in order to support multiple teaching practices. The implementation and the underlying calculations are described, focusing on the importance of learners' process data to provide teachers with a detailed and revealing view on their students' and class learning states. In an evaluation study of the dashboard's prototype with mock data, teachers (n=6) gave high ratings for the dashboard's usability.

CCS CONCEPTS

• **Applied computing** → **Interactive learning environments**; **Education**; • **Information systems** → **Data analytics**; • **Human-centered computing** → *Information visualization.*

KEYWORDS

teacher dashboard, learning analytics, intelligent tutoring systems, computer-assisted language learning

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1 INTRODUCTION

Teacher dashboards are applications of Learning Analytics (LA) that visualize information about learners and learner groups to support teachers. They are usually embedded in a learning technology system, such as Intelligent Tutoring Systems (ITS), that gathers data points from learners' learning products and processes. LA dashboards can then visualize aggregated data points in a timely manner and provide insights quickly [14]. In addition to the time-saving aspect, digital systems offer another significant advantage: they can efficiently collect data from all students as they engage with learning materials. This stands in contrast to a single teacher's limited ability to monitor the individual practice behavior and other indicators of the learning process of every student in the classroom. However, to maximize the benefits of data visualizations, it is important that they are understandable and meet the needs of teachers in order to be actionable [26].

The school context, especially in the area of language learning, is underrepresented in the literature, with a substantial portion of studies stemming from higher education [37] or focus on STEM subjects [e.g., 13, 45]. As studies demonstrate the benefits of ITS for students in English classes [24], the interpretability of data for teachers is of high importance for the practicality of these systems in the school environment.

We therefore formulate our research questions as follows:

RQ 1: Which functionalities do language teachers in K-12 classrooms request from a teacher dashboard?
RQ 2: Which data from an Intelligent Language Tutoring System (ILTS) with scaffolded feedback can be used in which way to provide metrics fulfilling the functions envisioned?
RQ 3: What is the dashboard's usability?

In this paper, we first provide a short overview of common challenges in dashboard design in the school context as well as our strategies for addressing them. Next, we describe our dashboard development process, which consisted of three steps: 1) a teacher survey study for functionality prioritization and needs analysis to address our first research question, 2) the development and implementation of the resulting teacher dashboard to address the second research question and 3) a simultaneous usability evaluation of the prototypical dashboard version with mock data to evaluate the

dashboard in terms of user experience to answer the third research question. We highlight expositions that make our dashboard different to previous work and point out potential limitations and future directions.

2 ADDRESSING THE RESEARCH GAP

With the rise of AI and machine learning based technologies, there is a growing presence of advanced digital learning tools, like ITS, to enhance students' self-regulated learning processes and learning outcomes. LA is a data-driven method for collecting and analyzing educational data from these tools to gain insights into the learning process and the performance of students [11]. While most LA-based educational tools are designed to interact directly with students and assist them in their individual learning progress, the role of teachers is to facilitate and support their students' (self-regulated) learning experiences [20]. LA can support teachers to make informed pedagogical decisions by providing additional information about their students' learning process beyond the regular interaction in class or performance records [16]. These insights are often presented through teacher-facing LA dashboards, which visually represent and aggregate the data generated by advanced learning systems for teaching purposes [37].

The scientific exploration of teacher dashboards is a relatively emerging field. Predominantly, LA research has centered on students, emphasizing the design and impact of student-facing dashboards (e.g. [3, 40]). While recent reviews indicate a noticeable increase in documented teacher-facing dashboard development in the literature, they also underscore certain issues and opportunities for enhancement [16, 43].

For instance, the majority of teacher dashboards is limited on exploratory projects and basic performance measures, neglecting theoretical aspects of self-regulatory learning and teachers' understanding of their students' cognitive processes [43]. However, a deeper understanding of the student learning process supports teachers to address the individual needs of their students, apply appropriate teaching strategies, and identify potential learning obstacles. This enables teachers to make their instruction more effective and optimize their students' learning experiences, ultimately leading to better learning outcomes [21]. LA dashboards yet rarely incorporate such process-oriented data [38], with a few noticeable exceptions, such as Dourado et al. [8], who however predominantly use click-stream data for analyzing learning processes. Since teachers use the information from LA dashboards as a source for their pedagogical decision-making [26], teacher-facing dashboards should be actionable, meaning that the information presented should directly enable teachers to make informed choices regarding appropriate pedagogical interventions.

Another recurring problem with newly developed technologies in education is their lack of practical application. To achieve sustained use of teacher dashboards in practice, user acceptance is fundamental [15]. Therefore, it is necessary to involve teachers in the entire design and creation process [7]. Although it is becoming increasingly common to include teachers into the development process, in most cases these efforts do not go beyond the exploratory phase of problem definition [16].

A related issue in teacher-facing dashboard development concerns information overload, which occurs when the demands of information processing exceed the capacity of available time [35]. In a study by van Leeuwen and colleagues [41], teachers often consulted the dashboard overview, but did not spend much time to revise detailed data. Therefore, it is necessary to provide the most important data at first sight, e.g., in an overview on the landing page, and to generally design teacher-facing dashboards with a clear structure, making information gathering easy and actionable.

In this paper, we describe the implementation of a teacher-facing LA dashboard for K-12 English teachers, which is based on an ILTS for students, i.e. the FeedBook [24, 32], from which the dashboard gets its data. The ILTS system features several exercise types, ranging from jumbled sentence to short answer exercises, and provides scaffolding feedback while working on an exercise [33] as well as criterion-referenced performance feedback on exercise submissions [6]. Additionally, ways of expanding the system with LA dashboard components have been explored [34] with an emphasis on student-facing LA. Some work on teacher-facing LA has been conducted, but not as holistic as in our present paper. Recently, the system has been extended with more user-adaptive features, incorporating macro-adaptive exercise selection, micro-adaptive exercise adjustments and an updated version of the task-oriented student dashboard [5].

Regarding the corresponding teacher-facing dashboard, we address the above mentioned issues and make several contributions to teacher dashboard development:

(1) Provide teachers with insights into students' learning processes. The underlying ILTS offers students scaffolding feedback concerning different error types. We leverage this feedback to provide teachers with comprehensive insights, encompassing not only exercise submissions and error frequency but also their students' progress in the learning process. Consequently, we make both class-wide and individual student error types accessible to the teacher, along with insights into acquired competencies and engagement metrics of the students.

(2) Incorporate a teacher-centered design. We employed a comprehensive co-design process involving teachers, which entailed incorporating their perspectives not only during the initial problem definition phase but also in the decision-making process for prioritizing dashboard features. Moreover, during the whole process of designing the different LA views that support the desired functionalities, we sought feedback from teachers regarding display and granularity of information. This approach enhances the direct practical utility of these functionalities from the teachers' standpoint, thereby increasing the likelihood of their effective application in the school context.

(3) Provide actionable, accessible data on the first view to foster internal differentiation and class orchestration. While student dashboards primarily focus on enhancing self-regulated learning, teacher dashboards should prioritize the effective utilization of student data to inform pedagogical interventions. A well-designed teacher dashboard should empower educators with the capability to directly address the unique needs of individual students and the collective needs of the entire class. As a solution, we have developed an efficient LA Quick View that consolidates essential information

into a single interface, allowing for a flexible focus on either individual students or the class as a whole. Where more information is needed, teachers can drill deeper and explore the other extended views. This design supports a variety of pedagogical actions and interventions.

In the following sections, we first describe the process and design of functionality prioritization by teachers and then show their implementation into the teacher dashboard and an evaluation of the resulting dashboard's usability.

3 FUNCTIONALITY PRIORITIZATION

To identify desired functionalities for a teacher dashboard, we conducted a needs analysis involving of the target group. To build a teacher dashboard that supports the teachers in their daily work, it needs to be actionable and tailored towards the teachers' needs [26]. Given varying occupational requirements for teachers at different educational levels, like school systems, school forms, grade levels or subject domains, we focus on the target group of the surrounding ILTS, i.e. seventh grade English teaching and learning in academic track secondary schools in Germany.

3.1 Participants and Procedure of the Functionality Prioritization

To identify the most important functionalities, we conducted a questionnaire survey asking 11 seventh grade English teachers from academic track schools in Germany to evaluate functionalities according to their importance for their daily work. These teachers were familiar with the non-adaptive version of the ILTS FeedBook (see Section 2), which they used with their seventh grade in English for approximately 12 weeks throughout an entire school year [28]. The version of the FeedBook used however did not incorporate a teachers dashboard but displayed only a list of individual exercise submissions ordered by submission time.

The participants were asked to rate a list of functionalities (see Table 1) in terms of their importance on a four-point scale ranging from "essential" (value 1), "useful", "not mandatory" to "unnecessary" (value 4), plus the additional category "cannot assess". The functionalities proposed were based on a review of existing dashboards and related publications, interviews with experts from language didactic, second language acquisition and teacher trainers [18]. As teachers might require different information when planning individual or group interventions, the questionnaire required the teachers to rank the functionalities on two dimensions: importance for class aggregated data, importance for student aggregated data. The entire questionnaire was asked in German language.

3.2 Results of Functionality Prioritization

To be actionable, the dashboard needs to be easy to use, and the functionalities need to be easy to understand. Moreover, the 'perfect' dashboard should be a time-saver and address users' common obstacles. One of them is the need for more in depth knowledge of individual learning states. This information is usually hard to obtain for every student in a timely manner. Dashboards that collect, aggregate and display data to teachers can be a great help in this regard. Regarding the desired functionalities, all teachers said that they would highly benefit from an aggregated view of the data collected in the ILTS. None of the teachers selected the category "cannot assess" for any of the functionalities. The functionality prioritization results, detailed in Table 1, guided the design and implementation of the dashboard presented in Section 4.

Display of strength, weaknesses and support needs was among the most desired functionality for student view (M=1.55, SD=0.522) and class view (M=1.73, SD=0.647). While an overview of common misconceptions shared first place with the former mentioned display of features desired in the class view (M=1.73, SD=0.647), it was not in the top three priorities in the students view but still rated as very important by the majority of the participating teachers (i.e., seven teachers assigned this functionality the value "1 (essential)"; M=1.55, SD=0.820). The highest rated functionality for the student dashboard was the overview of already acquired linguistic competencies (M=1.45, SD=0.522) which also scored high results for the class related priorities (M=1.82, SD=0.751). Suggestions of meaningful exercises was considered important on student (M=1.55, SD=0.522) as well as class level (M=1.82, SD=0.603). Overall, the four functionalities receiving the highest prioritization means for visualizations of individual student data are the same for the class-based view, varying only in the rank based on the means and standard deviations.

4 TEACHER DASHBOARD DEVELOPMENT

We designed the teacher dashboard based on several empirical findings, i.e. 1) the above described prioritization study, 2) the results of an expert design thinking workshop (including four teachers) and 3) qualitative interview studies with practitioners (n=15) [18]. The participating teachers highlighted the importance of overseeing both individual student data and aggregated class data. Further, they expressed the need for a quick overview as well as more detailed data for different use cases (preparation before class vs. feedback session with students or parents). We therefore designed a multi-page teacher dashboard incorporating views to display data relevant to support the desired functionalities, to answer our second research question. In addition, the whole design and development process involved iterative feedback loops with the five teachers involved in the project. We have taken into account teachers' assessed priorities in the dashboard, with one exception: Suggestions of meaningful practice exercises is not solved via the dashboard but rather via the macro-adaptive exercise sequencing mechanism within the ILTS.

The low-level data structure that nourishes all calculations is an attempt, a unique trial of a student on a single item. While students work with learning material (i.e., the exercises), they enter answers for exercise items. Immediately after, students receive information on their correctness and - if an answer is incorrect - additional scaffolding feedback, most in form of meta-linguistic cues. This feedback is enriched with linguistic annotations of correct and diagnosed linguistic constructs. Exercises themselves are pedagogically annotated with the topic they are tailored to practice and acquire through working on the exercises. Exercise meta-data is further enriched with linguistic annotations based on the exercises' textual content. This information of annotated student and target answers, correctness information and exercise meta-data together form the attempt, the baseline for all calculations in the teacher dashboard. Multiple artificial intelligence enhanced technologies are used in

	student view		class view	
Potential functionalities	M	SD	M	SD
Overview of already acquired linguistic competences	1.45	0.522	1.82	0.751
Overview of the processing status of exercises/tasks	1.64	0.505	2.27	0.905
Overview of grammatical error types (misconceptions)	1.55	0.820	1.73	0.647
Display of processing time and duration	2.55	0.688	2.55	0.688
Display of individual learning paths	2.82	0.874	2.82	0.982
Suggestions of meaningful exercises/tasks based on the learning data of the respective student(s)	1.55	0.522	1.82	0.603
Warning system	2.09	0.701	2.36	0.924
Display of strengths, weaknesses and support needs	1.55	0.522	1.73	0.647

Table 1: Showing mean (M) and standard deviation (SD) prioritization values for possible functionalities of an ILTS teacher dashboard rated by teachers (n=11) on a four-point scale from 1 "essential" to 4 "unnecessary".

the processing of learning material [30], student answers and feedback generation [33]. For the teacher dashboard we did not not use any additional machine learning or artificial intelligence (AI) methods to build the calculations, this makes it possible for teachers to understand (and even replicate if time allows) the system's calculations and results. "Black-box" systems in contrast are often missing transparency and may cause a lack of trust in K-12 teachers [27]. Particularly because machine learning models predictions of educational tasks are not always accurate enough and do not always overlap with teachers evaluation [10]. First steps towards unfolding the "black-box" have begun with the provision of explainable AI methods [19], but these still lack sufficient experimental validation when applied to educational settings [39].

The purpose of our teacher dashboard is to enable and support teachers, by providing them with a "window" to their students learning product and process data. Teachers can combine such information from the ILTS with their own classroom observations and knowledge about their students for educational intervention, which remains within the teacher's control.

4.1 Dashboard structure

We opted for a multi-page dashboard design to provide all functionalities but not overload a single display screen and thus the user. Moreover, teachers require different information for different instructional actions, thus we provide different views tailored for this purpose. The dashboard is structured as follows:

- Landing page with analytics quick view
- Class view with competencies, misconceptions and exercise completion sub-views
- Student view with competencies, misconceptions and exercise completion sub-views as well as an entry point to a read-only view of the students' dashboard

The landing page of the dashboard acts as an overview page providing the teachers with the most relevant information at a glance. The other pages can be accessed when teachers want to drill down into a specific content bundle for either the entire class or an individual student. Such a division into class and student views can facilitate decision-making for different instructional actions. For ease of understanding by teachers and intuitive navigation through the interface, the class and the student view are structured in a parallel way: the same information can be accessed at different aggregation levels, whether it is the data of an individual student within a selected language objective or the average of the whole class within that learning objective.

4.2 Dashboard implementation

The final teacher dashboard is embedded in the surrounding ILTS which contains additional non-data related functionalities such as user management (e.g., adding students, changing student passwords, adjusting settings) or learning content management (e.g., assembling learning units, viewing exercises, content locking and unlocking).

Throughout the entire iterative dashboard development process, as part of an interdisciplinary project, feedback from teachers was sought in keeping with a participatory design approach. Thus, the chosen visualization technique (e.g., pie chart, bar chart, matrix) is based on iterative discussions with the target users. As teacher seek explanations of different functionalities and to guide and support teachers while working with the dashboard, each view incorporates an information icon, displayed as 'i' icon in the top right corner, where detailed descriptions of the data displayed, and the underlying calculation can be found. This should foster two goals: 1) better understanding of information and thus enhance reasoning and make the data actionable; 2) trust in and acceptance of the system and its underlying data processing. In addition, following research-based design patterns, the color scheme is chosen to reflect relations across pages [2], for instance, mastery is consistently represented in an intense dark green.

In terms of learning content, teachers can create learning units tailored to their teaching practice or textbook by assembling different learning targets from a pool of options the system offers (e.g.,

"Simple Past"). Each learning target contains more fine-grained topics (e.g., "positive statements", "negative statements") which the teacher can include or exclude from the learning unit. These topics can be seen as ways of realizing the language target in spoken or written prose. Teachers can lock and unlock the included learning targets and topics for their class.

4.3 Landing page

The landing page, see Figure 1, is the start view in the teacher dashboard. This view was designed to meet teachers' need for a place to find information about strengths, weaknesses, and support needs as individuals and as a class in plenary as identified in the prioritization (see Chapter 3.2). Teachers can access it from within the ILTS FeedBook. If teachers have multiple classes, they are presented with a list of all their classes and select one before continuing to the dashboard landing page. This view offers most important information in one view, displayed in six different widgets providing information on different dimensions of learning and practice: progress, support needs, mastery, risk of forgetting, effort and improvement. These widgets were chosen based on iterative discussion with teachers involved in the design process. We asked them about their definition of strengths, weaknesses and support needs. Strength corresponded to mastery and progress, weakness corresponded to the lack of mastery or lack of progress - opposite ends of the same scale. Support needs were defined through the occurrence of frequent errors. Some teachers additionally also mentioned motivational support. This lead to the initial set of four widgets: progress, error-based support needs, mastery and effort. We additionally, after seeking approval from practitioners, added the risk of forgetting and improvement widgets to account for more theory-inspired features as proposed by the experts in the design thinking workshop.

Each widget exists in two variants, a 'students' and a 'topics' variant, providing a ranked list of either students or topics with respect to the learning dimension the widget represents. The division into these two variants is motivated by the fact that pedagogical interventions commonly target either individuals or the whole class and such a split can provide data aggregated on the level of the targeted intervention population. Matching the instrument to the professional activity is of great importance for its usefulness [25]. Moreover, teachers have different individual needs and preferences regarding information and their insights for pedagogical actions, which require different information [18]. Even though the effects of ranking lists, i.e., leaderboards, have been controversially discussed when presented to students [1, 22], they can bear useful insights when presented to teachers. During our interviews and discussions with practitioners, they explicitly mentioned that knowing the top and bottom three students or topics per category would be desired. Thus, we opted for ranked lists in the landing page widgets.

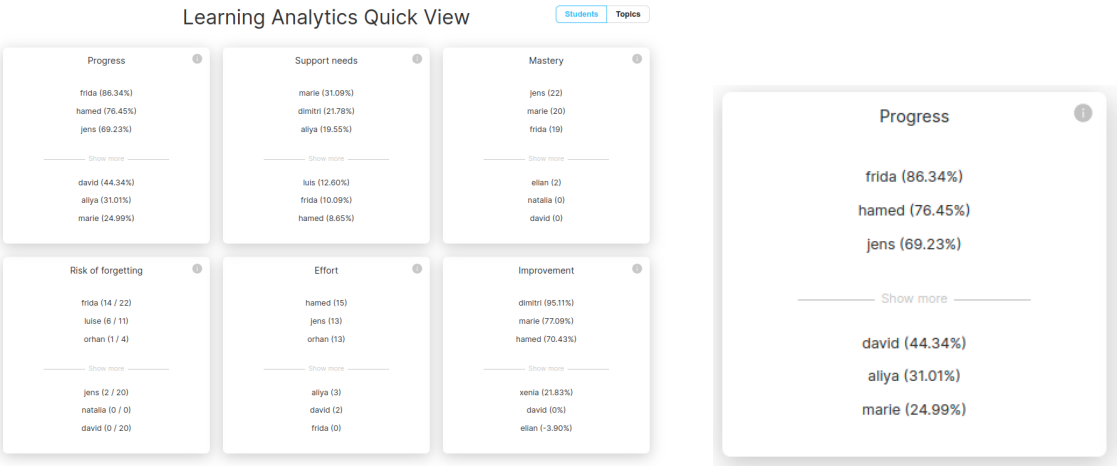
Besides the two fixed variants, teachers can choose to customize their own landing page variant, selecting a mixture of student or class centered widgets from the available selection. Thus, teachers have the opportunity to match the dashboard landing page to their personal preferences. While some widgets simply count occurrences and order the student or topics based on occurrence rations, other widgets, however, are based on more complex calculations. The ranked entities (students or topics) in the lists are accompanied by total or average counts to increase interpretability and comparison of results, especially in terms of divergence of unique data points. For all widgets on the landing page, all students' interactions with learning content that is available, i.e., unlocked, for the students for practice is considered for the calculations, they are not filtered for a specific learning unit or time frame. The landing page always represents the current overall state, if learning unit or topic specific information is searched, teachers can navigate to the respective view and filter the content accordingly.

4.3.1 Progress widget. Progress calculations in adaptive system - which include adaptive exercise sequencing and therefore do not have a set exercise sequence - can be challenging. In our system, progress can thus be seen as discrete steps. For each topic the students have to acquire a set of properties used in the topic's context (e.g., 'irregular verbs' in the context of the simple past). The number of properties varies between topics and each property acquired represents one discrete step towards topic mastery. The system's inherent adaptivity algorithm defines and evaluates property mastery [5]. Even though properties can be acquired in any order and do not follow a sequential order, steps can be seen as levels (e.g., 'level 3' corresponds to three mastered properties). Thus, the acquired properties of two students on 'level 2' for the same topic may differ. The progress widget calculation is based on these property acquisition steps. In the topic-centered view the progress value for a topic is based on the percentage mean progress level of all students. In the student-centered view the progress value for a student is based in the percentage mean progress level over all topics.

Based on learners' interactions with the system and the annotated properties, the system systematically updates the inherent learner models of the subject domain (English) across all activities. In this way, practice in one learning scope can result in learner model updates in other learning units, learning targets and realizations as well.

4.3.2 Support needs widget. Potential problems and thus support needs can be identified via incorrect attempts. Each practice exercise in the system is associated with a linguistic topic as the primary pedagogical goal, thus incorrect attempts can be associated with a linguistic topic. This widget displays ranked list of students or topics based on normalized ratios of incorrect to correct attempts.

4.3.3 Mastery widget. Topic mastery in the FeedBook can be seen as "readiness" in terms of having reached the necessary level of complexity for a topic as required by the learning unit's goal. Teachers select a suitable diagnostic exercise from a pool. When students answer this exercise (or an exercise of equivalent complexity, which targets the same linguistics constructions) and reach a predefined threshold without system support, i.e., no scaffolding feedback, the topic counts as mastered. Acquired mastery is visualized in the student dashboard as trophy [5]. Property mastery, as described in Section 4.3.1, can be seen as steps towards topic mastery. The landing page mastery widget thus ranks either topics or students by the number of trophies collected. Topics that have been mastered but are currently at risk of being forgotten (see Section 4.3.4) are still counted towards acquired trophies regardless of their forgetting state.



(a) Landing page with six widgets **(b) Progress widget (student-centered)**

Figure 1: Landing page (student-centered mode) of teacher dashboard with learning analytics quick view providing information about progress, support needs, mastery, risk of forgetting, effort and improvement

4.3.4 Risk of forgetting widget. The risk of forgetting is determined by the adaptivity algorithm. If students do not actively or passively practice topics or underlying properties for a while, these topics get flagged as 'potentially forgotten' and students need to successfully practice exercises including these properties in the exercise material to reactivate the trophy and thus prove the persistence of knowledge [5]. For the forgetting widget the ratio of forgotten to acquired trophies is calculated. Forgetting is a fundamental concept in learning [9]. Wisely placed repetition can overcome forgetting and support skill acquisition [17]. This widget is designed to make this salient to instructor to tailor their intervention accordingly if need be.

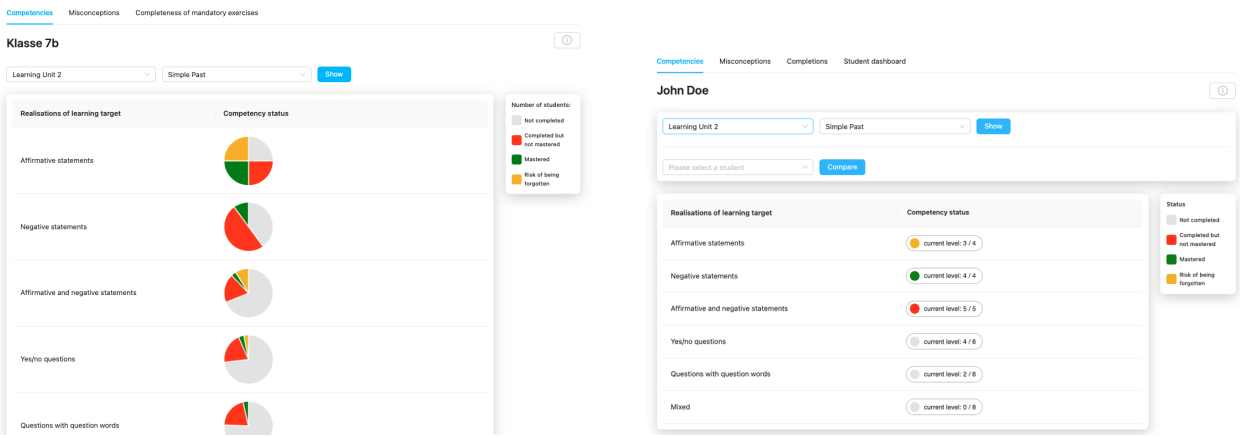
4.3.5 Effort widget. The effort widget is based solely on submission counts. The more exercises submitted, the higher the entry in the list. The raw count is given in brackets next to the students' or topics' name. Some students might not reach many mastery trophies but their effort rank could prove them hard working to the teacher and in this way, they can also contribute significantly to the overall topic-centered and thus class effort metric. Submissions with all items left unattempted are not counted as submission towards the effort criterion, thus cheating and just clicking submit on as many exercises as possible will not result in a higher effort rank.

4.3.6 Improvement widget. The improvement widget represents the intra-personal development. Improvement scores are calculated based on uptake ratios. Uptake is defined as correct attempt after an incorrect attempt. The uptake ratios (in percentage ranging from 0 to 100%) of two succeeding two-week intervals of an individual student are compared and the difference is taken as the improvement value for the student, result in a value range from -100% to +100%. For topic aggregated values, the sum of ratios of all students is normalized by the students count. Negative values indicate a decline in improvement, i.e., worsening, positive values indicate a true improvement and zero indicates no change. A value of zero could also be an early warning sign of students being stuck, this hypothesis however needs empirical validation, as being stuck might be multi-dimensional and a constant improvement rate trajectory with no slope (difference always equals zero) with very high performance could also indicate a stable performance (e.g., when students reach the mastery plateau and make no more errors). Succeeding weeks with uptake rations of zero, if no exercises are submitted, may also indicated topics loosing focus, which might then in return correlate with rising forgetting and is thus important to bring to teachers awareness.

The improvement widget serves multiple purposes. Besides informing teachers of students' uptake behavior, it also offers the possibility to put attention on those students whose performance is not outstanding, neither best nor worst. Praise, when applied correctly, is a promising factor to enhance students engagement [4, 12]. Teachers mention that praising is especially challenging for students that usually 'fly under the radar' [18]. By comparing individual to themselves and not to peers, the class midfield as well as students who lag behind the class average or are even underperforming can be encouraged for any improvement.

4.4 Competencies view

Besides the landing page, the dashboard also offers additional views. The first one is the competencies view, see Figure 2, which was considered most wanted by the teachers regarding the student view (see Section 3.2). In this view, teachers can filter by learning unit and learning target. For the selected scope of learning material, the underlying grammatical topics are displayed. The ability to see competence status at the level of fine-grained grammatical subunits enables teachers to provide more targeted instruction and support for students' self-regulated learning. An example of such a competence is "Yes/No questions". Mastery for this competence can be defined as the students ability of producing grammatically valid sentence structures that are of type question and can be answered



(a) Class view **(b) Student view**

Figure 2: Competencies view illustrating the competency status for a selected learning scope, in the example the different grammatical topics that are assembled under 'Simple Past' in the 'Learning Unit 2'

with the binary response "yes" or "no" (e.g., "Did you go to the store yesterday?").

In the class view, for each topic a pie chart is displayed, showing the distribution of competency states among the students of the class, each student contributing one piece to the pie. A piece can take on one of the following values representing the mastery status:

- **Not completed** (gray): The student has not yet reached the last level, which corresponds to having shown evidence for achieving all required properties.
- **Completed but not mastered** (red): The student achieved all properties for the topic, but did not reach the required threshold in the diagnostic exercise (or did not attempt the diagnostic exercise).
- **Mastered** (green): The student reached mastery by showing the required performance in the diagnostic exercise.
- **Risk of forgetting** (yellow): After having mastered the topic, the student has not shown active or passive knowledge of all underlying properties for a hiatus period.

In the student view, a full circle per topic is shown, colored according to the mastery status. Next to the circle the current level of properties the student has reached and the maximum level for the topic is indicated as text (e.g., "level 4/6"). Regarding forgetting, if a student has mastered a topic previously but the hiatus has passed for one of six properties, the circle will be colored in yellow indicating "risk of forgetting" and the text label will indicate "level 5/6" as "level 6", the sixth property, has to be re-activated.

As teachers often seek an easy way to compare two students, the view offers to select a second student whose competencies circles and labels for the learning target are displayed next to those of the previously selected student.

4.5 Misconceptions view

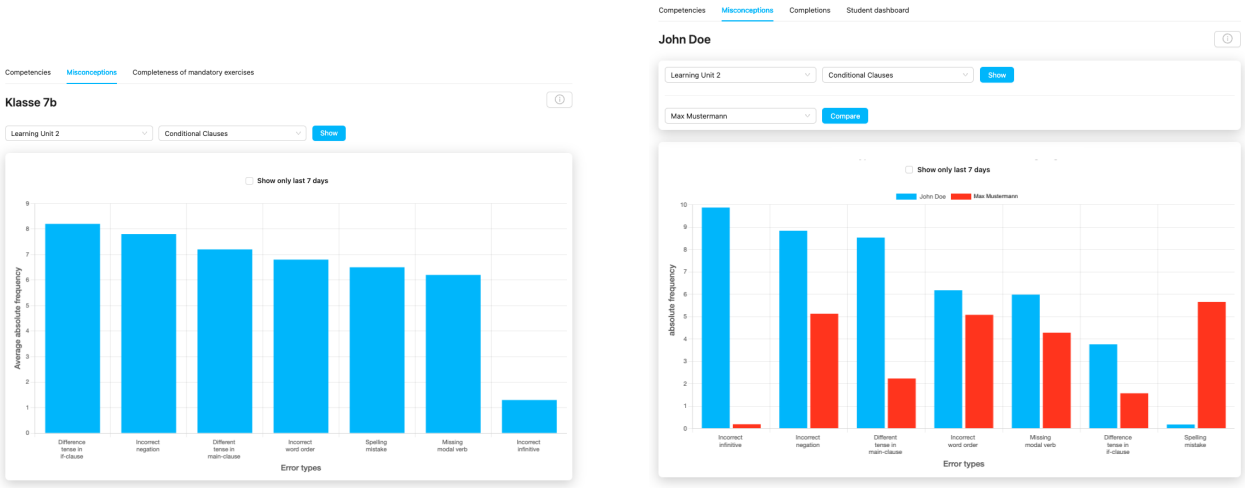
The misconceptions view, see Figure 3, focuses on the incorrect attempts and the related error types of students while working through exercises. It was among the top two for class view and fourth place for student view in our prioritization study (see Section 3.2). The meta-linguistic feedback provided for an incorrect attempt combines information of expected and provided construct, such as "With conditional clauses (type 2), we use the simple past in the if-clause, not the simple present" or "Simple past with regular verbs: verb + -ed. For verbs ending in consonant + 'y', change the 'y' to 'i'". These different feedback messages and annotations carry information about the error types. They are counted and visualized in a histogram. The comparison feature to contrast two students is available in this view as well, see Figure 3b.

As in the competencies view, teachers can use the filter to select a learning unit and a learning target, only those exercises are considered to calculate the error frequencies. To enable teachers to focus on recent development of students, this view can be filtered to consider only the latest five submissions per student to the learning content scope, i.e. selected learning unit and learning target. Thus, initial errors are not dragged along but can be mitigated over time if they do not reoccur. Each incorrect attempt, raw data point, can be seen as an error, but errors not necessarily correspond to misconceptions, they could also be careless mistakes or oversights. Misconceptions are systematic errors that represent erroneous presumptions about the topic or a specific rule. The misconceptions view showing counts of errors per student can work as a mean to help identify and differentiate mistakes and misconceptions.

4.6 Exercise completion view

Not only the learning state is relevant for teachers, but also exercise completions play an important role, especially when evaluating homework submissions. Class aggregated values were not rated as particularly important by the study participants, but for the student view, even though not being ranked as high as the other functionalities, all teachers selected either 1 "essential" or 2 "useful". Hence, we decided to include an exercise completion view, see Figure 4, but backed away from providing cross-student averages.

Because the system provides students with adaptively sequenced practice exercise, not all students may receive the same exercises.



(a) Class view

(b) Student view with active student comparison

Figure 3: Misconceptions view representing the error occurrences and types, a filter to only see errors conducted in the last seven submissions per student and learning target can be activated

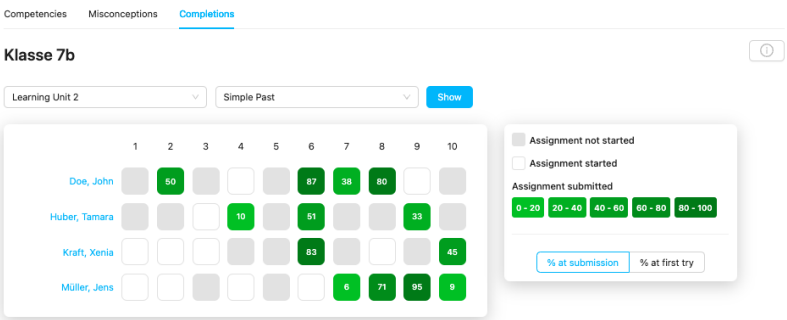


Figure 4: Class view for mandatory exercises with values in cells displaying submission accuracy in percentage as selected via legend

However, the system provides the opportunity of mandatory exercises per learning target, which are exercises that at learning unit creation are selected by the teacher and can be assigned to the whole class. These exercises are accessible also outside of the adaptive sequence (for more detail see [5]). For these exercises, submissions can be accessed in the exercise completion view.

The completion view for mandatory exercises renders a matrix of submissions with rows representing students, only one row in the student view, and columns representing distinct mandatory exercises. For each submission, a cell in the matrix, the submission status (not started, started, submitted) and for submitted exercises the accuracy values is displayed as color highlighted number in the cell, the better the accuracy the darker the green. In the legend, teachers can toggle a button to switch between seeing accuracies at first try, without feedback, or at submission time (i.e., correct at first try + correct after feedback). All other practice exercise submissions, assembled and selected by the adaptivity algorithm, can be accessed via the teachers' read access on the student dashboard. The student dashboard for a particular student can be accessed from the navigation panel within the student view.

Due to privacy concerns, which are often neglected in LA dashboard development [16, 44], we actively decided not to disclose submission timestamps to teachers. Additionally, the number of attempts as well as individual attempts themselves are also not accessible for the teachers to respect students' privacy and to allow students to enter as many unseen attempts as desired before finally hitting the submit button, hence releasing and sharing only the last attempt per item with the teacher, i.e., answers teachers would also see in case of paper-based submissions.

5 USABILITY EVALUATION

After we finished the dashboards' design and during the implementation phase, we conducted a usability study with an interactive, clickable prototype with mock data to answer to assess usability and answer our third research question before implementing the

dashboard in real school settings. Thus, this prototype was created prior to the final implementation, and some usability optimizations resulting from the usability evaluation were incorporated into the final implementation which has been described in the previous section (Section 4).

5.1 Participants and Procedure of Usability Evaluation

The usability study was conducted with six English teachers (n=6). At first, the participants were introduced to the FeedBook system by a test administrator via video conference. Next, they were asked to click trough the dashboard prototype using the think aloud methodology [42], i.e. verbalizing their understanding of the functionalities, impressions, irritations, or questions about the dashboard. Thereafter, they completed an online questionnaire assessing usability and preferences on specific details in displays (e.g., percentages or absolute numbers). We used the German version of the User Experience Questionnaire (UEQ) to assess the dashboard usability [23]. The UEQ contains 26 items in the form of semantic differentials with a seven-point scale, which can be grouped in accordance with six factors: Attractiveness, Efficiency, Perspicuity, Dependability, Stimulation, and Novelty.

For the usability study, we used an interactive clickable prototype with mock data, not yet a full integrated version of the dashboard. The version of the dashboard presented in Section 4 differs from the prototype used in the usability evaluation in the following points, which the users were informed about by the test administrator:

- The prototype was not integrated with the FeedBook system, thus system functionalities regarding content and user management could not be explored.
- Default options for filters were pre-selected (e.g., learning unit and learning target).
- "Forgetting" was not taken into account for the prototype and thus not included in the landing page and the competencies view for the prototype.
- The landing page in the prototype contained a "customized landing" consisting of five selected widgets, two topic-centered (progress, support needs) and three student-centered (effort, mastery, improvement).

Moreover, there were some aspects optimized for the final implementation, as described in Section 4, based on the teachers thinking aloud and survey input:

- The final implementation contain explanation buttons to get information on each view.
- The compare option in the student view for competencies and misconceptions was added after the prototype.
- In the completeness of exercises view, the percentages on submission and first try were added.

5.2 Results of Usability Evaluation

We used the UEQ Data Analysis Tool [1] to analyze the data [31]. The results as displayed in Figure 5a indicate overall high usability, all six factors have positive mean values. These scores indicate a good overall user experience, which can be backed up by the the results from the think aloud part. The highest value was obtained for stimulation, which shows the teachers' high motivation to use the dashboard. However, the standard deviation for some of the scales, especially attractiveness, are rather large indicating the need for a larger sample size to evaluate these factors accurately.

The UEQ Data Analysis Tool also offers a benchmark evaluation where current results are compared with the results of previous collected product ratings [2] [36]. Overall, our dashboard achieved scores above average on all scales, see Figure 5b.

6 DISCUSSION AND CONCLUSION

Teacher dashboards are promising information sources for teachers working with ILTS. But to really benefit the teachers - by providing additional information and simultaneously be a time saver - they must be designed comprehensibly, be aligned with teachers' needs and be actionable [26]. We therefore conducted a needs analysis to identify which information teachers demand to support and facilitate their decision-making process (see Section 3). Results showed that teachers placed high emphasis on overviews regarding competencies and error types, as well as information on strengths, weaknesses and support needs, both on class and individual level. They also wished for suggestions of meaningful exercises. Based on these results we designed and implemented a teacher dashboard for English teachers of students in secondary schools in Section 4. The clickable prototype of the completed design, but prior to the final implementation, reached high ratings in our usability survey. In contrast to the majority of teacher dashboards, the dashboard presented incorporates not only product data or click behavior but also includes process data of practice episodes that is linguistically annotated and thus enriched with information from the learning domain, in our case English language. This has the advantage that teachers can gain insights into deeper cognitive processes of their students, not just the outcome of the interaction with the system. Additionally, with its landing page that can display topic- or individual-centered metrics and the structural split into student and class aggregated views, the dashboard is designed to support teachers when performing different pedagogical actions. With the widgets, teachers can quickly and specifically access information about students' learning progress and that of the class. Furthermore, they can gain deeper insights into views related to competencies and error types. The division into class and individual data aligns with the typical use cases for teachers in the school context.

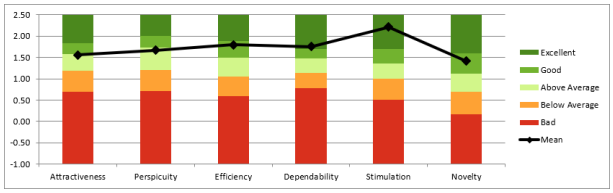
We acknowledge that the design, however, may have some limitations. The dashboard comprises multiple views and the landing page contains multiple widgets, which might lead to information overload in teachers with low data literacy skills. To mitigate this risk a user study using eye tracking methodology like the method proposed in [29] could provide clues as to where the user experience can be improved. Although the dashboard is based on a needs analysis survey, the interactive prototype has been positively evaluated in a small-scale usability evaluation and the implementation has been conducted in a participatory setting with several teachers, the system should be further evaluated in authentic settings with respect to both usability and effectiveness. Such an evaluation could

[1] https://www.ueq-online.org/

[2] Benchmark dataset downloaded in April 2024, containing data from 21175 persons on 468 comparison product evaluations.

UEQ Scales (Mean and Variance)		
Attractiveness	1.556	3.61
Perspicuity	1.667	0.97
Efficiency	1.792	0.54
Dependability	1.750	0.73
Stimulation	2.208	0.54
Novelty	1.417	1.62

(a) UEQ mean and variance



(b) UEQ benchmark comparison

Figure 5: Results of UEQ evaluation

determine if the high rating from the small-scale usability evaluation, i.e., the evaluation conducted using the clickable prototype with mock data, can be replicated with a larger sample size and if it transfers to real-life teaching and learning. Such an evaluation is currently ongoing as part of a large-scale randomized controlled field trial [3]. Moreover, the widgets on the landing page are subject to continuous improvement based on teacher feedback and the pool of options to select during landing page customization will be extended in future iterations. For instance, a more elaborate version of the effort widget incorporating also time-on-task is planned for a future version. Additionally, extensions of the dashboard could be implemented. Currently the system incorporates monitoring functionalities, presenting the data collected in an accessible way, but one could imagine adding alerting and advising functionalities.

The development of LA technologies and their application in teacher-facing dashboards is advancing rapidly. Especially in the school context, where sensitive student data is involved, and the consequences of intransparent or unintelligible design could have serious consequences, scientifically-based development is essential. This paper provides initial guidance on how to systematically integrate teachers' needs into the development process and how a teacher dashboard can provide valuable and actionable information in a clear and concise dashboard structure.

REFERENCES

[1] Fernando RH Andrade, Riichiro Mizoguchi, and Seiji Isotani. 2016. The bright and dark sides of gamification. In *Intelligent Tutoring Systems: 13th International Conference, ITS 2016, Zagreb, Croatia, June 7-10, 2016. Proceedings 13*. Springer, 176–186.

[2] Benjamin Bach, Euan Freeman, Alfie Abdul-Rahman, Cagatay Turkay, Saiful Khan, Yulei Fan, and Min Chen. 2022. Dashboard design patterns. *IEEE Transactions on Visualization and Computer Graphics* 29, 1 (2022), 342–352.

[3] Robert Bodily and Katrien Verbert. 2017. Review of research on student-facing learning analytics dashboards and educational recommender systems. *IEEE Transactions on Learning Technologies* 10, 4 (2017), 405–418.

[4] Paul Caldarella, Ross AA Larsen, Leslie Williams, Kade R Downs, Howard P Wills, and Joseph H Wehby. 2020. Effects of teachers' praise-to-reprimand ratios on elementary students' on-task behaviour. *Educational Psychology* 40, 10 (2020), 1306–1322.

[5] Leona Colling, Tanja Heck, and Detmar Meurers. 2023. Reconciling Adaptivity and Task Orientation in the Student Dashboard of an Intelligent Language Tutoring System. In *Proceedings of the 18th Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2023)*. 288–299.

[6] Leona Colling, Ines Pieronczyk, Cora Parrisius, Heiko Holz, Stephen Bodnar, Florian Nuxoll, and Detmar Meurers. 2024. Towards Task-Oriented ICALL: A Criterion-Referenced Learner Dashboard Organising Digital Practice. In *Proceedings of the 16th International Conference on Computer Supported Education - Volume 1: CSEDU*. 668–679. https://doi.org/10.5220/0012753000003693

[3]The trial is conducted in the realm of the AI2Teach-Project (https://uni-tuebingen.de/de/256625) using the dashboard described in the paper and embedded in the FeedBook (https://ai2teach.de/feedbook/)

[7] Mollie Dollinger, Danny Liu, Natasha Arthars, and Jason M Lodge. 2019. Working together in learning analytics towards the co-creation of value. *Journal of Learning Analytics* 6, 2 (2019), 10–26.

[8] Raphael A Dourado, Rodrigo Lins Rodrigues, Nivan Ferreira, Rafael Ferreira Mello, Alex Sandro Gomes, and Katrien Verbert. 2021. A teacher-facing learning analytics dashboard for process-oriented feedback in online learning. In *LAK21: 11th International Learning Analytics and Knowledge Conference*. 482–489.

[9] Hermann Ebbinghaus. 1885. *Über das gedächtnis: untersuchungen zur experimentellen psychologie*. Duncker & Humblot.

[10] Roberta Galici, Tanja Käser, Gianni Fenu, and Mirko Marras. 2023. How close are predictive models to teachers in detecting learners at risk?. In *Proceedings of the 31st ACM Conference on User Modeling, Adaptation and Personalization*. 135–145.

[11] Carolina Guzmán-Valenzuela, Carolina Gómez-González, Andrés Rojas-Murphy Tagle, and Alejandro Lorca-Vyhmeister. 2021. Learning analytics in higher education: a preponderance of analytics but very little learning? *International Journal of Educational Technology in Higher Education* 18 (2021). https://api.semanticscholar.org/CorpusID:233865710

[12] Matthew R Hodgman. 2015. Student praise in the modern classroom: The use of praise notes as a productive motivational tool. *Journal of Education and Training* 2, 1 (2015), 41–47.

[13] Nicole Hutchins and Gautam Biswas. 2023. Using Teacher Dashboards to Customize Lesson Plans for a Problem-Based, Middle School STEM Curriculum. In *LAK23: 13th International Learning Analytics and Knowledge Conference*. 324–332.

[14] Ioana Jivet, Maren Scheffel, Marcus Specht, and Hendrik Drachsler. 2018. License to evaluate: Preparing learning analytics dashboards for educational practice. In *Proceedings of the 8th international conference on learning analytics and knowledge*. 31–40.

[15] Rogers Kaliisa, Anna Gillespie, Christothea Herodotou, Anders Kluge, and Bart Rienties. 2021. Teachers' perspectives on the promises, needs and challenges of learning analytics dashboards: Insights from institutions offering blended and distance learning. In *Visualizations and dashboards for learning analytics*. Springer, 351–370.

[16] Rogers Kaliisa, Ioana Jivet, and Paul Prinsloo. 2023. A checklist to guide the planning, designing, implementation, and evaluation of learning analytics dashboards. *International Journal of Educational Technology in Higher Education* 20, 1 (2023), 28.

[17] Sean HK Kang. 2016. Spaced repetition promotes efficient and effective learning: Policy implications for instruction. *Policy Insights from the Behavioral and Brain Sciences* 3, 1 (2016), 12–19.

[18] Mareike Kholin, Leona Colling, Nina Lamberti, and Detmar Meurers. in preparation. Elevating K-12 Education: Innovations in Learning Analytics Teacher-Facing Dashboards and Evidence-Based Best Practices. (in preparation).

[19] Hassan Khosravi, Simon Buckingham Shum, Guanliang Chen, Cristina Conati, Yi-Shan Tsai, Judy Kay, Simon Knight, Roberto Martinez-Maldonado, Shazia Sadiq, and Dragan Gašević. 2022. Explainable artificial intelligence in education. *Computers and Education: Artificial Intelligence* 3 (2022), 100074.

[20] Bracha Kramarski. 2018. Teachers as agents in promoting students' srl and performance: applications for teachers' dual-role training program. In *Handbook of self-regulation and of learning and performance*. Routledge, 223–239.

[21] Mareike Kunter and Silvia Ewald. 2016. Bedingungen und Effekte von Unterricht: Aktuelle Forschungsperspektiven aus der pädagogischen Psychologie. *Bedingungen und Effekte guten Unterrichts* 1 (2016).

[22] Richard N Landers and Amy K Landers. 2014. An empirical test of the theory of gamified learning: The effect of leaderboards on time-on-task and academic performance. *Simulation & Gaming* 45, 6 (2014), 769–785.

[23] Bettina Laugwitz, Theo Held, and Martin Schrepp. 2008. Construction and evaluation of a user experience questionnaire. In *HCI and Usability for Education and Work: 4th Symposium of the Workgroup Human-Computer Interaction and Usability Engineering of the Austrian Computer Society, USAB 2008, Graz, Austria, November 20-21, 2008. Proceedings 4*. Springer, 63–76.

[24] Detmar Meurers, Kordula De Kuthy, Florian Nuxoll, Björn Rudzewitz, and Ramon Ziai. 2019. Scaling up intervention studies to investigate real-life foreign language learning in school. *Annual Review of Applied Linguistics* 39 (2019), 161–188.
[25] Inge Molenaar and Carolien Knoop-van Campen. 2017. Teacher dashboards in practice: Usage and impact. In *Data Driven Approaches in Digital Education: 12th European Conference on Technology Enhanced Learning, EC-TEL 2017, Tallinn, Estonia, September 12–15, 2017, Proceedings 12*. Springer, 125–138.
[26] Inge Molenaar and Carolien AN Knoop-van Campen. 2018. How teachers make dashboard information actionable. *IEEE Transactions on Learning Technologies* 12, 3 (2018), 347–355.
[27] Tanya Nazaretsky, Mutlu Cukurova, and Giora Alexandron. 2022. An instrument for measuring teachers' trust in AI-based educational technology. In *LAK22: 12th international learning analytics and knowledge conference*. 56–66.
[28] Cora Parrisius, Ines Pieronczyk, Carolyn Blume, Katharina Wendebourg, Diana Pili-Moss, Mirjam Assmann, Sabine Beilharz, Stephen Bodnar, Leona Colling, Heiko Holz, et al. 2022. Using an intelligent tutoring system within a task-based learning approach in English as a foreign language classes to foster motivation and learning outcome (Interact4School): Pre-registration of the study design. (2022).
[29] Stanislav Pozdniakov, Roberto Martinez-Maldonado, Yi-Shan Tsai, Namrata Srivastava, Yuchen Liu, and Dragan Gasevic. 2023. Single or Multi-page Learning Analytics Dashboards? Relationships Between Teachers' Cognitive Load and Visualisation Literacy. In *European Conference on Technology Enhanced Learning*. Springer, 339–355.
[30] Martí Quixal, Björn Rudzewitz, Elizabeth Bear, and Detmar Meurers. 2021. Automatic annotation of curricular language targets to enrich activity models and support both pedagogy and adaptive systems. In *Proceedings of the 10th Workshop on NLP for Computer Assisted Language Learning*. 15–27.
[31] Maria Rauschenberger, Martin Schrepp, Manuel Pérez Cota, Siegfried Olschner, and Jörg Thomaschewski. 2013. Efficient measurement of the user experience of interactive products. How to use the user experience questionnaire (UEQ). Example: Spanish language version. (2013).
[32] Björn Rudzewitz, Ramon Ziai, Kordula De Kuthy, and Detmar Meurers. 2017. Developing a web-based workbook for English supporting the interaction of students and teachers. In *Proceedings of the joint workshop on NLP for Computer Assisted Language Learning and NLP for Language Acquisition*. 36–46.
[33] Björn Rudzewitz, Ramon Ziai, Kordula De Kuthy, Verena Möller, Florian Nuxoll, and Detmar Meurers. 2018. Generating feedback for English foreign language exercises. In *Proceedings of the thirteenth workshop on innovative use of NLP for building educational applications*. 127–136.
[34] Björn Rudzewitz, Ramon Ziai, Florian Nuxoll, Kordula De Kuthy, and Detmar Meurers. 2019. Enhancing a Web-based Language Tutoring System with Learning Analytics.. In *EDM (Workshops)*. 1–7.
[35] Allen G. Schick, Lawrence A. Gordon, and S. Haka. 1990. Information overload: A temporal approach. *Accounting Organizations and Society* 15 (1990), 199–220. https://doi.org/10.1016/0361-3682(90)90005-F
[36] Martin Schrepp, Jörg Thomaschewski, and Andreas Hinderks. 2017. Construction of a benchmark for the user experience questionnaire (UEQ). (2017).
[37] Beat A Schwendimann, Maria Jesus Rodriguez-Triana, Andrii Vozniuk, Luis P Prieto, Mina Shirvani Boroujeni, Adrian Holzer, Denis Gillet, and Pierre Dillenbourg. 2016. Perceiving learning at a glance: A systematic literature review of learning dashboard research. *IEEE Transactions on Learning Technologies* 10, 1 (2016), 30–41.
[38] Gayane Sedrakyan, Jonna Malmberg, Katrien Verbert, Sanna Järvelä, and Paul A Kirschner. 2020. Linking learning behavior analytics and learning science concepts: Designing a learning analytics dashboard for feedback to support learning regulation. *Computers in Human Behavior* 107 (2020), 105512.
[39] Vinitra Swamy, Sijia Du, Mirko Marras, and Tanja Kaser. 2023. Trusting the explainers: teacher validation of explainable artificial intelligence for course design. In *LAK23: 13th International Learning Analytics and Knowledge Conference*. 345–356.
[40] Natercia Valle, Pavlo Antonenko, Kara Dawson, and Anne Corinne Huggins-Manley. 2021. Staying on target: A systematic literature review on learner-facing learning analytics dashboards. *British Journal of Educational Technology* 52, 4 (2021), 1724–1748.
[41] Anouschka Van Leeuwen and Nikol Rummel. 2020. Comparing teachers' use of mirroring and advising dashboards. In *Proceedings of the tenth international conference on learning analytics & knowledge*. 26–34.
[42] Maarten Van Someren, Yvonne F Barnard, and J Sandberg. 1994. The think aloud method: a practical approach to modelling cognitive. *London: AcademicPress* 11, 6 (1994).
[43] M Wiedbusch. 2021. Intelligent teacher dashboards that support students' self-regulated learning, engagement, and teachers' decision making. 6 (2021).
[44] Kimberly Williamson and Rene Kizilcec. 2022. A review of learning analytics dashboard research in higher education: Implications for justice, equity, diversity, and inclusion. In *LAK22: 12th international learning analytics and knowledge conference*. 260–270.
[45] Françeska Xhakaj, Vincent Aleven, and Bruce M McLaren. 2017. Effects of a teacher dashboard for an intelligent tutoring system on teacher knowledge, lesson planning, lessons and student learning. In *Data Driven Approaches in Digital Education: 12th European Conference on Technology Enhanced Learning, EC-TEL 2017, Tallinn, Estonia, September 12–15, 2017, Proceedings 12*. Springer, 315–329.